SHRIYA PRASANNA

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EDUCATION

University of Washington Seattle, Washington ⋄ *Graduation December 2027*
Bachelor's of Science in Physics
Bachelor's of Arts in Music, vocal and operatic performance

Selected Coursework: Graduate General Relativity I (PHYS 564), Graduate Galactic Structure & Dynamics (ASTR 511)

RESEARCH EXPERIENCE

University of British Columbia/TRIUMF – Nuclear Physics Researcher
June - 2026 to present
PI: Dr. Jason Holt (TRIUMF) *Vancouver, British Columbia*

- Calculating nuclear matrix elements (NME) for exotic decays.
- Using ab-initio nuclear theory to calculate higher-order corrections to NME's for double-beta decay.

***NSF REU* University of Alaska Fairbanks Department of Mathematics & Statistics – Fluid Dynamics Researcher**
June - 2025 to August- 2025
PI: Dr. Alexei Rybkin (UAF) *Fairbanks, Alaska*

- Derived an analytic solution to the inverse tsunami-wave problem, modeling propagation and inundation of tsunamis in the inverse-direction.
- Numerically verified solution by recovering initial profile of the wave from data at the shoreline.
- Utilized Mathematica and Python to derive solutions for the Nonlinear Shallow Water Equations (NSWE) and Boussinesq partial differential equation systems.
- Wrote a paper, which was accepted to Springer's *Water Waves*, see **PUBLICATIONS** section.

University of Washington Department of Astronomy – Milky Way Galaxy Dynamics Researcher
January - 2025 to present
PI's: Dr. Nora Shipp (UW), Dr. Arpit Arora (UW), Dr. Peter Ferguson (UW) *Seattle, Washington*

- Building a physically-motivated model of the Milky Way Galaxy potential using cosmological simulations and data.
- Modeling stellar stream dynamics in the time-evolving Milky Way system.
- Quantifying orbital uncertainties of stellar streams in evolving vs. static potentials, challenging traditional stream science methods.
- Modeling the orbits of stellar streams by fitting simulations to data; using a variety of statistical inference tools.
- Utilizing MCMC, HMC, and GPU-accelerated statistical methods.

Saint Martin's University Department of Physics & Astronomy – Observational Astronomy Summer Student
July - 2024 to November- 2024
PI: Dr. Andrea Kunder (SMU) *Lacey, Washington*

- Studied low-luminosity globular clusters in the Milky Way (MW) Bulge.
- Characterized low-luminosity candidate globular clusters using IRAF spectra reduction software.
- Calculated radial velocities of cluster member stars.
- Used data from the VVV Survey and Gaia proper motions.
- Presented poster at Murdock conference.
- **1 Night of Observing at SOAR Telescope**.

PUBLICATIONS

1. Obrobnikov O., Jones M., **Prasanna S.**, et al. (2026) "Reconstruction of the Non-linear Wave at a Buoy from Shoreline Data and Applications to the Tsunami Inverse Problem for Piece-Wise Sloping Bathymetry." *Water Waves*, arXiv:2510.14177

OBSERVING EXPERIENCE

- **SOAR Telescope**: Cerro Pachón, Chile – 4.1 Meter Telescope 1 Night
  Remotely took spectra for 9 different Milky Way bulge globular cluster candidates.
- **Višnjan Observatory**: Tićan, Croatia – 1 Meter Telescope 8 Nights
  NEO follow-up, see selected publications below from the MPEC website for candidate $L01$:
  Measurer: September 8, 2026
  Observer: September 7, 2026

TECHNICAL SKILLS :

Programming Languages: Python, Mathematica, IRAF, Astronomical Data Query Language (ADQL)
Libraries & **Software:** Jax, Agama (Action-based galaxy modelling architecture), Git/Github, Slurm

AWARDS

June 2026: DiRAC Institute Summer Research Prize, Seattle, WA

- Research award given to outstanding UW physics & astronomy undergraduates to do research in the UW astronomy department

September 2025: Hans Wolf Fellows Award, Seattle Washington

- Scholarship awarded to one outstanding vocal performance student at the UW School of Music.

June 2025: NSF REU Grant, Fairbanks, Alaska

- NSF grant to fund my Reseach Experience for Undergraduates (REU), a prestigious award for undergraduate researchers.

PRESENTATIONS & CONFERENCES

Probing the Orbital History of Streams in Time-Dependent Acceleration Fields *(Poster)*
UW Department of Astronomy Undergraduate Research Symposium, Seattle, WA, ***October 2026***

Near-Field Cosmology in the Era of Big Data *(Invited Conference)*
The Fields Institute, University of Toronto, Toronto, ON, ***July 2026***

Times Up! How Time-Evolution of the Milky Way Affects Stellar Streams *(Talk)*
Mary Gates Undergraduate Research Symposium, University of Washington, Seattle, WA, ***May 2026***

Spectroscopy of FSR 1775 – Is it a Globular Cluster? *(Poster)*
J. Murdock Undergraduate Research Symposium, Vancouver, WA, ***November 2024***

REFERENCES

Dr. Nora Shipp, University of Washington, Department of Astronomy, Seattle, WA
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Dr. Arpit Arora, University of Washington, Department of Astronomy, Seattle, WA
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Dr. Alexei Rybkin, University of Alaska, Department of Mathematics & Statistics, Fairbanks, AK
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Dr. Andrea Kunder, St. Martin's University, Department of Physics & Astronomy, Lacey, WA
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